

— STARTING SHORTLY —

RAG: giving agents a *brain* of your own data.

Lumière Bakery has three branches, eight sections of knowledge, and one problem: customers keep asking the same questions. Today you build the agent that answers them — from Lumière's own documents, not from a guess.

You walked in with a confident guesser. You leave with an agent that cites its source.

GUESSER – NO DOCUMENT LOADED

Q: "Can I get a 2kg eggless chocolate cake for this Saturday?"

A: *For a 2kg eggless chocolate cake, you typically need 48–72 hours' notice. Lumière usually charges around ₹2,200 for that size. Standard 50% advance applies.* △ Fabricated – no document was ever loaded.

GROUNDNED – LUMIERE_KB.MD LOADED

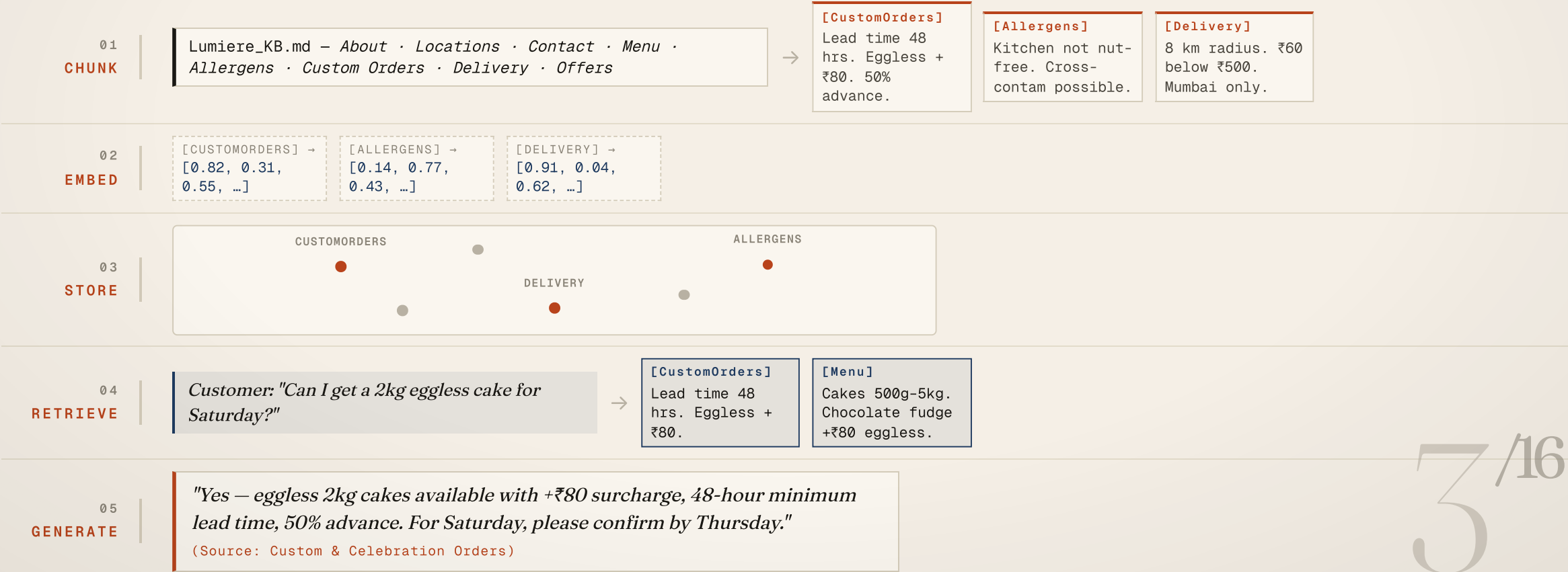
Q: "Can I get a 2kg eggless chocolate cake for this Saturday?"

A: *Yes — 2kg cakes are available. Eggless option is +₹80. Minimum 48 hours' advance notice required for any custom cake. 50% advance to confirm.*

(Source: Custom & Celebration Orders – Lead Times)

— FIVE STEPS, ONE DOCUMENT

Chunk · embed · store · retrieve · *generate.*



— THREE CUSTOMER QUERIES · TWO RETRIEVAL LANES

It matches *meaning*, not the exact words.

"no-egg option for cakes?"

KEYWORD ✗ "no-egg" doesn't appear in the doc.

SEMANTIC ✓ Matches "eggless" by meaning.

"deliver near Versova?"

KEYWORD ✗ "Versova" sits inside an address line
— wrong context for retrieval.

SEMANTIC ✓ Finds the Andheri West delivery
zone.

"pincode 400058 deliverable?"

KEYWORD ✓ Exact digit match on the Andheri
zone.

SEMANTIC ~ May find nearby zones — the digits
aren't "meaning."

*Different blind spots in each lane. **Hybrid** runs both at once — meaning catches the phrasing, keyword catches the codes. That's Build E.*

— A MAP OF MEANING · LUMIÈRE'S CHUNKS PLOTTED

Where the pieces actually live: a *vector database*.



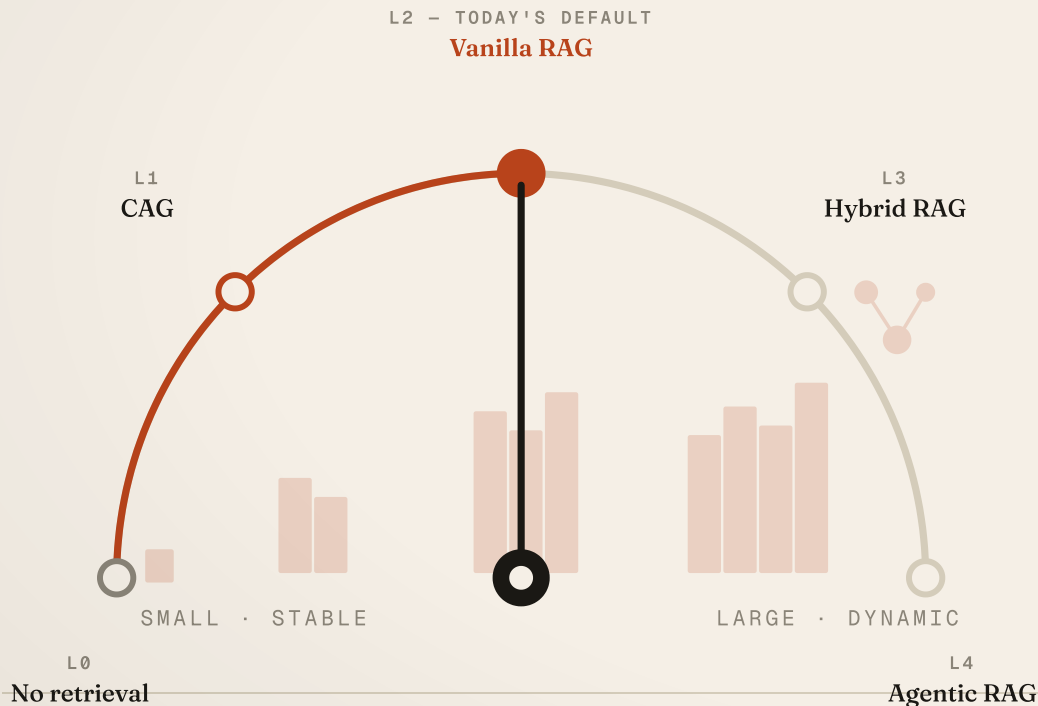
MAP OF MEANING
Each chunk is a point.
Similar ideas sit close.
"Find me similar" becomes
"find me nearby."

THE FORK — MANAGED OR
SELF-HOST
Pinecone — managed;
someone runs the machine.
Weaviate — self-host;
your team runs it.

WHY IT SCALES
Today, one doc loads
instantly. At 500 stores
and a 10,000-page manual,
this map is what makes
retrieval still fast.

— TYPES OF RAG · TURN IT UP AS THE KNOWLEDGE GETS BIGGER

RAG is a *dial*, not a switch.



L0	No retrieval	Ask the cake question with nothing loaded → it guesses a lead time. The baseline.
L1 · Build A	CAG	Load whole Lumiere_KB.md into context. No vector DB. Small, stable knowledge.
L2 · B-D	Vanilla RAG	500 locations, 10,000-page manual — retrieve only the matching sections. ~80% of cases.
L3 · Build E	Hybrid RAG	Keyword + semantic fire together. "Clause 3" and "eggless cake" both land.
L4 · S5	Agentic RAG	The agent decides when to retrieve, across multiple sources. Concept today.

WORDS YOU'LL HEAR, NOT TODAY'S BUILD — **GraphRAG:** retrieves over a map of how entities connect — for cross-document synthesis, where plain RAG is for precise single-fact lookup. · **Re-ranking RAG:** retrieve twenty pieces, a second model re-sorts to the best three before answering — for precision-critical work.

BUILD A · FEED THE BRAIN · ~8 MIN

A

Give the agent its brain.

Create a Project, upload Lumiere_KB.md, ask one question from your persona — and watch it answer from the file. Chunk, embed, store, all behind the glass. This is CAG, level one on the dial.

PROMPT

```
Create a Project named "Lumière Knowledge Agent."  
Add Lumiere_KB.md to Project Knowledge.  
Ask one question from your persona's Build A query.  
→ Confirm the answer pulled a real detail from the file.  
  
Fallback: fresh chat + Lumiere_KB.md attached – same result.
```

BUILD B · MAKE IT CITE · ~8 MIN

B

Make it show its work.

One short instruction turns "trust me" into "here's where I found it." Every answer comes back with a receipt — the section it pulled from.

PROMPT

Add to your Project instructions:

"You are a knowledge agent for Lumière Bakery. Answer questions using ONLY the document(s) in your knowledge. After every answer, name the section, heading, or page you pulled it from, like this: (Source: <section/heading>). Do not add outside facts."

→ Re-ask. Confirm it ends with a real (Source: ...) line.

BUILD C · MAKE IT REFUSE · ~4 MIN



Teach it to say "I don't know."

Ask something your doc doesn't cover — watch it guess. Then add one line. Now it refuses. The boundary between a grounded answer and an honest refusal is the difference between a demo and something safe to deploy. Especially on allergens — never guess.

PROMPT

First, ask your persona's out-of-scope question. Watch it guess.

Then add to your instructions:

"If the answer is not in the document, say: 'I don't have that in my knowledge base' – and stop. Never guess, never fill in from outside knowledge."

→ Re-ask. Confirm it now refuses instead of guessing.

- BREAK -

10:00

Stand up. Water. Stretch.

GROUNDED, CITED, HONEST. NEXT WE MAKE IT STURDY.

10¹⁶

BUILD D · TUNE THE RECALL · ~12 MIN

D

Break it on purpose, then make it findable.

The agent is only as good as the document you feed it. Today's demo: a wall-of-text policy doc that breaks retrieval, then the same content structured with headings and tags — and the query lands. The model didn't get smarter. The document got findable.

PROMPT

Watch the facilitator demo:

- MESSY policy doc: ask a cancellation-rules query.
Note what the agent gets wrong or misses.
- CLEAN policy doc: same query, structured doc.
Note how it lands and cites correctly.

Fill in your KB Design Sheet:

- What chunk strategy did the clean doc use?
- What section tags made the difference?

Then test your own Lumière agent:

- Ask a question you think it should answer.
- If it misses — add a heading or tag to that section, re-upload.

11/16

12:00

BUILD E · STRESS & SHIP · ~14 MIN

E

Three probes, then finalize.

The last test is the real world being awkward — wrong words, adjacent questions, exact codes. Attack it three ways, then lock and ship.

PROMPT

Synonym probe – ask in different words than the doc uses.

→ "no-egg option" when the doc says "eggless."

→ Confirm semantic still finds it.

Out-of-scope probe – ask something adjacent but absent.

→ Confirm the refusal still holds.

Exact-code probe – ask for a pincode, clause, or section number.

→ "What pincodes does the Andheri branch cover?"

→ Keyword fires on the code; meaning handles the prose: hybrid.

Lock grounding + citation + refusal. Shipped.

12¹⁶

14:00

— FOUR SHAPES · ONE PIPELINE

The same five steps, *four jobs*. Find yours.

Document Q&A

CONSULTING · LEGAL · HEALTHCARE

Point it at a contract, protocol, or report — and ask.

Corporate / policy bot

HR · OPS · CUSTOMER SERVICE

The handbook, the SOP, the same ten questions everyone asks — like Lumière's WhatsApp queue.

Internal knowledge search

FINANCE · OPS

Wikis, shared drives, the Confluence nobody can navigate.

Product knowledge base

MARKETING · SALES

Specs, pricing, the playbook — so reps stop guessing. Lumière's menu and delivery zones are exactly this.

Don't boil the ocean. Start with one high-impact function drowning in repeat questions. Prove the retrieval is accurate. Earn that team's trust. *Then* scale. The teams that fail tried to ingest the whole company on day one.

13¹⁶

— THE DECISION MATRIX · WHAT CHANGES VS. WHAT STAYS

RAG vs fine-tuning, in *one grid*.

	WHAT YOU'RE CHANGING → FACTS	WHAT YOU'RE CHANGING → BEHAVIOR
STATIC (RARELY CHANGES) ↑	EITHER WORKS Static facts Lumière's founding story, French-technique baking principles. RAG fits; fine-tune fits — pick the cheaper one.	FINE-TUNE Static behavior House voice. Always reply in the brand's warm-formal tone. JSON output every time. Train it once.
DYNAMIC (CHANGES OFTEN) ↓	RAG ← TODAY Dynamic facts This week's offers, new Gurgaon branch, allergen updates, delivery zones. Update the document — never retrain the model.	BOTH (HYBRID) Dynamic behavior Rare. A bot whose tone must adapt per customer. RAG for context, fine-tune for voice scaffolding.

One honesty line: fine-tuning does *not* remove being wrong — it makes the model wrong in your brand voice. Grounding is what gives an answer a source. And this rule is practitioner consensus, not settled law — expect honest debate at the edges.

— THE REAL PRIZE YOU WALK OUT WITH

The Knowledge Base *Design Sheet*.

The agent is today's demo. The sheet is the recipe — five fields you fill for the Lumière agent you just built, then reuse on any document at work.

★ **Where on the RAG dial?** One sentence — "mine is level-1 CAG / level-2 vanilla / level-3 hybrid" — and why.

01 **Chunk strategy** — how you'd split Lumiere_KB.md: by section, branch, or topic.

02a **Structural tags** — section labels so retrieval finds the right piece (e.g. [CustomOrders], [Allergens], [Delivery]).

02b **Semantic tags** — the question(s) each section answers (e.g. "Can I get a same-day cake?").

03 **Query range** — what this agent should answer, and what it must refuse (e.g. competitor pricing, revenue, GPS tracking).

04 **Grounding + refusal prompt** — your final custom-instructions text.

Not a smarter model. A *grounded* one.

Five builds — feed, cite, refuse, tune, ship. A Lumière Knowledge Agent that answers from the document, names its source, refuses what it doesn't know, and handles both plain questions and exact pincodes. And the recipe written down, so you do it again Monday on a document no one has seen.

— TAKE-HOME, TWO SENTENCES EACH —

One document at work you'll ground an agent on this week, and the first question you'll want it to answer. One point on the RAG dial your most common use-case lands on — and why.

— NEXT SESSION —

*Session 5 — **Agent Architecture: how agents think, remember & act.** Today you gave the agent a brain made of Lumière's documents; next we give it the will to use that brain on its own.*